Racial Biases in Hiring: The Effect of Race on Hiring Practices

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Introduction

In this study, we look at discrimination in hiring based on race. The goal is to show the effects of race on hiring and determine if race does affect one's ability to be employed. Discrimination is an ongoing issue in the United States. African Americans specifically have dealt with systemic discriminatory practices in almost every aspect of their lives. Realizing discrimination in hiring and working to fix it can broaden the talent pool, allowing more people to contribute their talents and skills. It also brings diverse perspectives, reducing likelihood of "group-think" that can hurt organizations. For many businesses, it is also about understanding the consumer. Having a workforce that reflects a business's consumer base allows them to better understand their market and align with their customers' goals.

In today's political climate, Diversity, Equity, and Inclusion (DEI) practices at companies, universities, and institutions nationally are being slashed. The current administration has taken aggressive action against DEI policies, labeling them as discriminatory, wasteful, and anti-meritocratic. Through executive orders and political influence, DEI programs in the government, government-affiliated companies, schools, universities, and contractors were cut after they faced losing federal funding for continuing DEI initiatives.

This study aims to show the effects of race on hiring practices. The main question is: Are white people more likely to be called back for a job than black people? With the results, the goal is to shed light on potential discrimination in hiring practices and emphasize the importance of DEI education and equal opportunity employment initiatives.

Null Hypothesis (H0): There is no significant effect of race on the likelihood of receiving a callback from a job.

Alternative Hypothesis (H1): There is a significant effect of race on the likelihood of receiving a callback from a job.

Data Description

This study will be using the dataset from [Bertrand & Mullainathan's (2004) Resume Experiment]. This data was collected and processed by Bertrand & Mullainathan. The dataset was obtained from [OpenIntro], an independent non-profit who specialize in high-quality and accessible educational tools for teachers and students. The link for this dataset and its metadata is: https://www.openintro.org/data/index.php?data=resume.

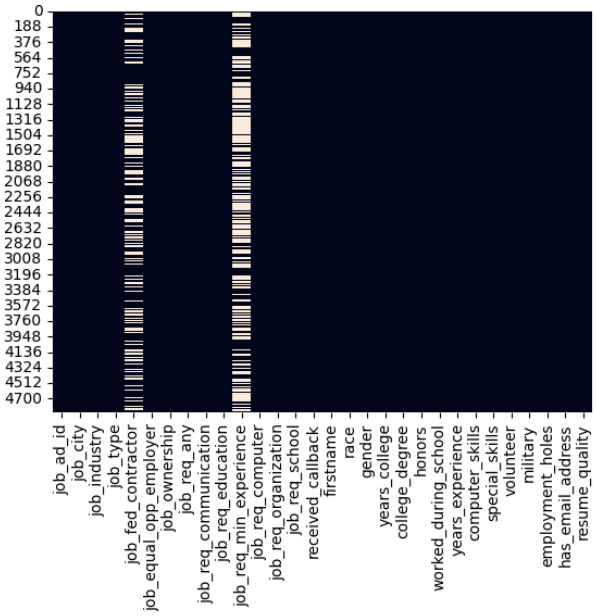
The dataset had 4870 observations (N=4870), and 30 variables (K=30) that describe applicant details, job details, and the outcome. There are 30 total variables, some of which are tied together, some to ensure randomness and variability, and descriptors of the applicants or jobs. Metadata, descriptors, and details can be found at the source link above.

	job_ad_id	job_req_min_experience	years_college	years_experience
count	4870.000000	3806.000000	4870.000000	4870.000000
mean	651.777823	0.811876	3.618480	7.842916
std	388.690698	1.584858	0.714997	5.044612
min	1.000000	0.000000	0.000000	1.000000
25%	306.250000	0.000000	3.000000	5.000000
50%	647.000000	0.000000	4.000000	6.000000
75%	979.750000	1.000000	4.000000	9.000000
max	1344.000000	10.000000	4.000000	44.000000

The summary statistics table shows the distribution of the data. One issue is that the data is largely binary, meaning a summary statistics table would not show much, other than the mean that would tell the proportion of 0/1 in the feature. The summary table shown has the binary data filtered out. The column job_req_min_experience also needed cleaning. I believed it was important to show this feature as it is one of the few continuous columns, but it included the value "some" which represented a

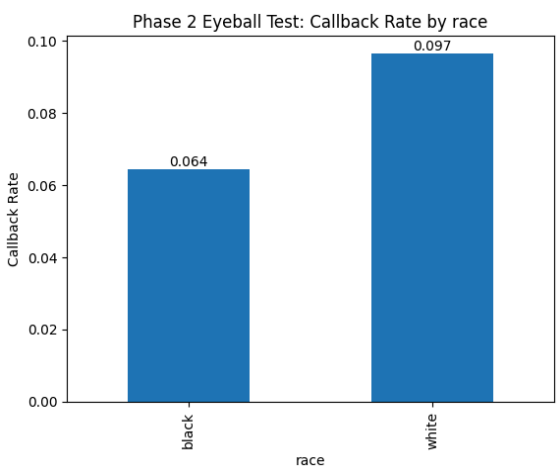
job posting claiming that some experience was required. I could not assign this an unbiased replacement, so I removed those instances from the summary table.

The data is extremely reliable, consistent, and relatively clean. In order to check the data for any missing values, I used a seaborn heatmap plot. This graph visually shows any gaps in the data allowing me to identify the completeness and any issues. Looking at the data from the descriptors, info, and heatmap, we can see that it is almost entirely complete with some missing data in two columns: job_fed_contractor and job_req_min_experience. The prevalence of the missing data and completeness of all the other fields led me to believe that the data was not really incomplete but structurally missing/MNAR (Missing Not at Random).



Taking a closer look at the raw dataframe and the meta data, I concluded that the missing data represents a "0" or "none" value. This would make sense for job_req_min_experience, which represents the amount of experience required for the job. I assume NaN values represent no experience needed, especially given its prevalence in the data. Although, the value of "0" is used too, so it is hard to confidently make that assumption. In the case of the column job_fed_contractor, which is the indicator for if the employer is a federal contractor, I have some issues. In the raw data for the file, there are values for "0" (no) or "1" (yes) already inputted. The values that came back as NaN are written as "NA" in the file. I cannot confidently claim what the "NA" values represent since a company can only be or not be a federal contractor. It could be not applicable to the specific job listing, but there would be no way of determining that since the job listings are anonymous. In both these cases, I will assume that the NaN is a "0" value, and the issues were just differences with the collection of the data.

An important note about the resume data is that it is entirely fictitious. All applicants are created by the original authors. While the resumes are created based on real profiles and templates, it does not contain any real information. Each applicant was given a resume and it was categorized into low quality and high quality. Then, four resumes, two high quality and two low quality were selected for each job posting. One high quality resume and one low quality resume was given a traditionally white name, while the other high and low quality resumes were given a traditionally black name. This gave no resume a bias towards a race, as race was randomly assigned to a resume. The real information is related to the job postings and the callback rate of the fictitious applicants. If you would like to know more about the creation of the data, you can find the methods in [section 3.1 and 3.2 of Bertrand & Mullainathan's paper]() in the original paper (pgs. 6-8). This means there are no outliers or issues in the data that we would find when working with real-world sampled data. This bar graph is the "Eyeball Test" conducted to see if there was a difference in callback rate by race. Given that the race is randomly assigned, we know that any difference between the two can only be attributed to discrimination in hiring. The important check is whether the difference is statistically significant.

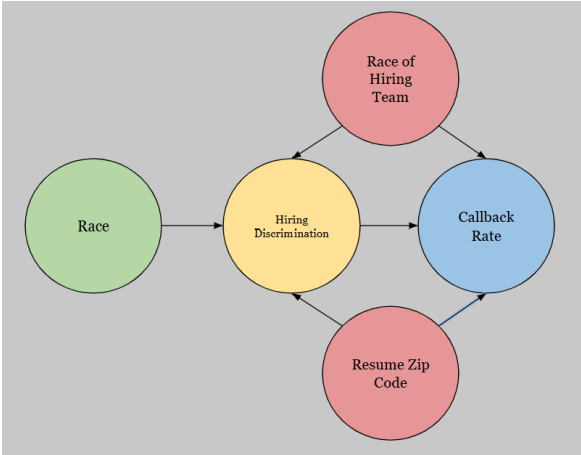


Methodology

When determining the methodology, it is important to keep in mind the original question: Are white people more likely to be called back for a job than black people?

The independent variable (D) is race. We are mainly testing the effect of "race" on an applicant's likelihood to receive a call back for a job. "Firstname" was chosen based on the selected race of the resume, but no other variable was affected by race. The resumes were also

unbiasedly assigned a race/name. The dependent variable (Y) is whether an applicant got called back after applying for a job ("received_callback"). This is what we will use to measure the effect of race on hiring. The control variables (X) that I will be focusing on are resume_quality, computer_skills, years_experience, and college_degree. I chose these variables because I believe they are important characteristics of a competitive resume and en. There are many other variables to consider, especially when discussing some of which may appear in the methods, findings, and conclusions. One variable that I eventually decided to veer away from is gender as it was considerably affected during collection.



When considering confounding variables, two were clear: Race of Hiring Team and Resume Zip Codes. Someone who is white will likely have a bias towards white names because of their familiarity with their "in-group" (people tend to prefer or choose familiar things). Given our data does not include information about the race of the hiring teams, we cannot correctly control for this. The paper discusses the zip code/neighborhood as another confounding variable. The resumes had randomized addresses, so the original authors tested the neighborhood's role in the hiring process. Our data does not include this. However, we cannot determine whether the choice of discrimination was because of the race or the perceived social class. With some black names being associated with lower socioeconomic background, a black name might signal more than just a racial bias.

The final equation I used for my model was:

'received_callback ~ race + resume_quality + computer_skills + years_experience + college_degree'

This equation regressed the callback rate on the above mentioned variables. The model I implemented this equation with was a probability unit or "probit" model. A probit model is a

regression analysis used to model binary outcomes with a normal distribution function. This model is used in economics research frequently, including in the original paper. A probit regression is the most appropriate choice because of its ability to understand binary outcomes, which in our case is callback (0/1, no/yes). The logic of the normal distribution function of a probit model applies to the desirability of each applicant. The assumption is that applicants' desirability by employers is normally distributed. The other option would be a logit regression. The main advantage of that would be the practicality of coefficients. Logit coefficients would show that being black impacts the callback odds by [coefficient]. Although this would be helpful, to keep consistent with the original paper and for the normal distribution, I continued with the probit model.

I also included two secondary probit models that differentiated by race. This is consistent with the original paper, and also provides interesting insights on how the tested features' effects in both groups. The equations were the same except without including race.

'received_callback ~ resume_quality + computer_skills + years_experience + college_degree'

The data was filtered by race when running the probit model

To account for heteroskedasticity, I set the covariance type to 'cluster' and grouped by 'job_ad_id'. Cluster covariance accounts for intragroup correlation, as opposed to the other options that are a general correction for heteroskedasticity. Recall that 4 applications were sent to the same job posting. This creates many groups within the data that the model should take into account.

Results

The primary probit model successfully ran with strong significance. The model is highly significant, with the predictors explaining the callback probability better than using the average callback rate for everyone. Looking at the coefficients, race, resume_quality, and year_experience showed significance, computer_skills was weakly significant, and

college_degree showed no significance. Having used a probit model, I also needed a marginal effect table to show the actual effects of the features. All interpretations are “ceteris paribus,” meaning that all other variables were held constant.

Being white increased the probability of receiving a callback by 3.14 percentage points. With the baseline callback rate (baseline is being black) at 6.4 percent, an increase by 3.14 percentage points is 49 percent. It also aligns with the original eyeball test, where the difference was about 3.3 percentage points. This was extremely significant with a p-value at near 0.

```
Optimization terminated successfully.
         Current function value: 0.275870
         Iterations 6
                          Probit Regression Results
==============================================================================
Dep. Variable:      received_callback   No. Observations:                 4870
Model:                         Probit   Df Residuals:                     4864
Method:                           MLE   Df Model:                            5
Date:                Wed, 25 Mar 2026   Pseudo R-squ.:                 0.01465
Time:                        03:21:46   Log-Likelihood:                -1343.5
converged:                       True   LL-Null:                       -1363.5
Covariance Type:              cluster   LLR p-value:                 1.532e-07
=========================================================================================
                            coef    std err          z      P>|z|      [0.025      0.975]
-----------------------------------------------------------------------------------------
Intercept                -1.4894      0.116    -12.862      0.000      -1.716      -1.262
race[T.white]             0.2129      0.042      5.103      0.000       0.131       0.295
resume_quality[T.low]    -0.1011      0.048     -2.109      0.035      -0.195      -0.007
computer_skills          -0.1363      0.075     -1.805      0.071      -0.284       0.012
years_experience          0.0184      0.006      3.331      0.001       0.008       0.029
college_degree           -0.0277      0.064     -0.435      0.664      -0.153       0.097
=========================================================================================

       Probit Marginal Effects
=====================================
Dep. Variable:      received_callback
Method:                          dydx
At:                           overall
=========================================================================================
                           dy/dx    std err          z      P>|z|      [0.025      0.975]
-----------------------------------------------------------------------------------------
race[T.white]             0.0314      0.006      5.029      0.000       0.019       0.044
resume_quality[T.low]    -0.0149      0.007     -2.109      0.035      -0.029      -0.001
computer_skills          -0.0201      0.011     -1.795      0.073      -0.042       0.002
years_experience          0.0027      0.001      3.306      0.001       0.001       0.004
college_degree           -0.0041      0.009     -0.435      0.664      -0.022       0.014
=========================================================================================
```

Having a low quality resume reduced callback probability by 1.49 percentage points. This makes sense, but also puts race’s effect into perspective. Being black hurts an applicant twice as much as having a low quality resume does.

Each additional year of experience increased callback probability by 0.27 percentage points. This effect does make sense, as an applicant with more experience would be more desirable. It also shows that a black applicant would need over 11 years of experience to close the gap between its white counterpart. Again, exemplifying how strong the effect of race is.

Computer skills and college degree attainment were not statistically significant. Although, it is worth noting the odd effects. Computer skills were negatively associated with the callback rate, and college degree was extremely insignificant.

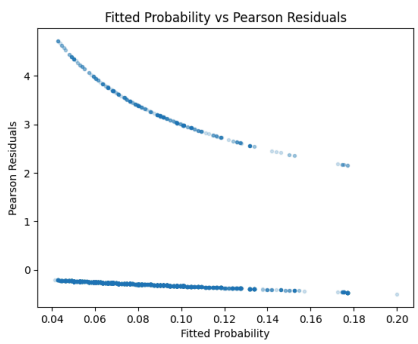
When running the probit models on the data for each race, we were able to see how variables affect each variable differently. It is difficult to compare the two as black applicants had almost no statistically significant effects other than years_experience. In both cases, each additional year of experience increased an applicant's probability of callback by 0.3 percentage points at a significant level. All other features were insignificant on black applicants. This shows that some credentials are almost invisible when looking at black applicants compared to white applicants. This follows a finding from the original paper, discrimination bites twice. Not only does being black decrease baseline chances for being called back, but it also reduces the validity of credentials.

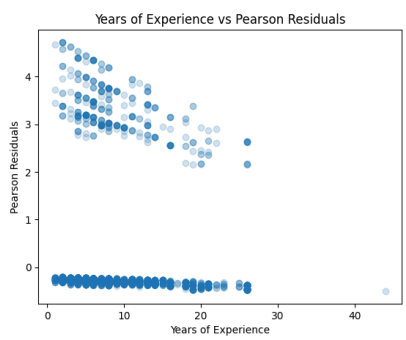
```
       Probit Marginal Effects - White Applicants
====================================
Dep. Variable:      received_callback
Method:                          dydx
At:                           overall
==============================================================================================
                            dy/dx    std err          z      P>|z|      [0.025      0.975]
----------------------------------------------------------------------------------------------
resume_quality[T.low]    -0.0275      0.010     -2.780      0.005      -0.047      -0.008
computer_skills          -0.0345      0.016     -2.188      0.029      -0.065      -0.004
years_experience          0.0028      0.001      2.408      0.016       0.001       0.005
college_degree           -0.0115      0.013     -0.881      0.378      -0.037       0.014
==============================================================================================

       Probit Marginal Effects - Black Applicants
====================================
Dep. Variable:      received_callback
Method:                          dydx
At:                           overall
==============================================================================================
                            dy/dx    std err          z      P>|z|      [0.025      0.975]
----------------------------------------------------------------------------------------------
resume_quality[T.low]    -0.0026      0.009     -0.305      0.760      -0.019       0.014
computer_skills          -0.0049      0.014     -0.357      0.721      -0.032       0.022
years_experience          0.0027      0.001      2.775      0.006       0.001       0.005
college_degree            0.0032      0.011      0.295      0.768      -0.018       0.025
==============================================================================================
```

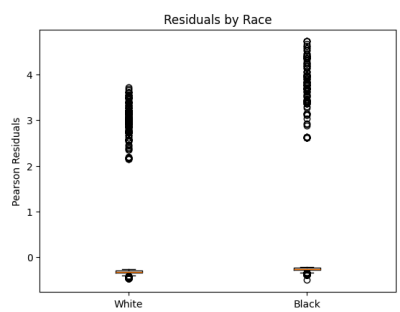
Robustness Checks

To ensure the model is robust, I decided to test whether the treatment of race stays consistent across different model formulas. To test this, I ran a probit model several times, adding a control variable and comparing the coefficient of race and its p-value each time. Across each model, the coefficient range was 0.005 and maintained extreme significance. This was an extremely small shift, showing no omitted variable bias and that race is not correlation with any other variable. This lines up with the data's creation/collection as no race was attached to the attributes of the resumes when they were created. With the race treatment maintaining significance across all models, we can see the discrimination finding is not a result of any resume characteristic.

I also plotted the Pearson residuals. Unlike OLS regressions, probit regressions do not produce the standard residuals. The Pearson residual shows the difference between observed and fitted values and can help identify outliers. It shows how well the models fit, and with the binary result, it will show the consistency across the treatment of race. The first plot of the residuals against the probability shows the models prediction abilities across the different probabilities. The model does not show over or under predicting at different probabilities. Next I plotted the residuals against years of experience. Year of experience is the only continuous variable in the original formula, but more importantly it was consistent in the probit models that were separated by race. With no noticeable pattern in the residuals, we can be sure there are no missing terms that would highly affect the model. Finally, the test of residuals in each race shows that the model predicts accurately in both groups. While there are some higher outliers with black resumes, the median, Q1, and Q3 are all close to 0.







Conclusion

I reject the original null hypothesis, with significant evidence showing that race affects the likelihood of receiving a callback from a job meaning the alternative hypothesis is likely true. The primary probit model shows that being white increases your probability of receiving a callback from a job by 49% with extreme statistical significance. All models show strong

significance and good results towards the main treatment. I found no evidence of missing variables causing unreliable results, with the model passing robustness tests.

There are many limitations and assumptions of the study. One major limitation was the data collected. It was collected from an online library, which although reliable, did not contain all the data from the original study. Being able to test for some omitted variables, like the effect of zip code, would provide a stronger understanding of the true effect of race by allowing us to add control variables. There are also many variables that cannot be collected, like the race of the hiring team. Having this variable would, again, allow us to control for bias that went undetected. Another limitation is that the applicants are not real, so we cannot test for the effect of race through the whole hiring process. With many jobs having in-person interviews, some with multiple rounds, not being able to test the effects of discrimination through interviews also creates a limitation.

These findings can be used to further DEI policies and equal opportunity employment procedures. Not only does it shine light on discrimination in hiring, but also the discrimination in resume credentials. With black applicants being bitten twice, by initial discrimination but their resume credential not helping them in the same way it would a white applicant. This is important to consider when implementing policy and programs. Training programs that target black people will not close the gap. Instead, it is a systemic issue that must be targeted at its roots.